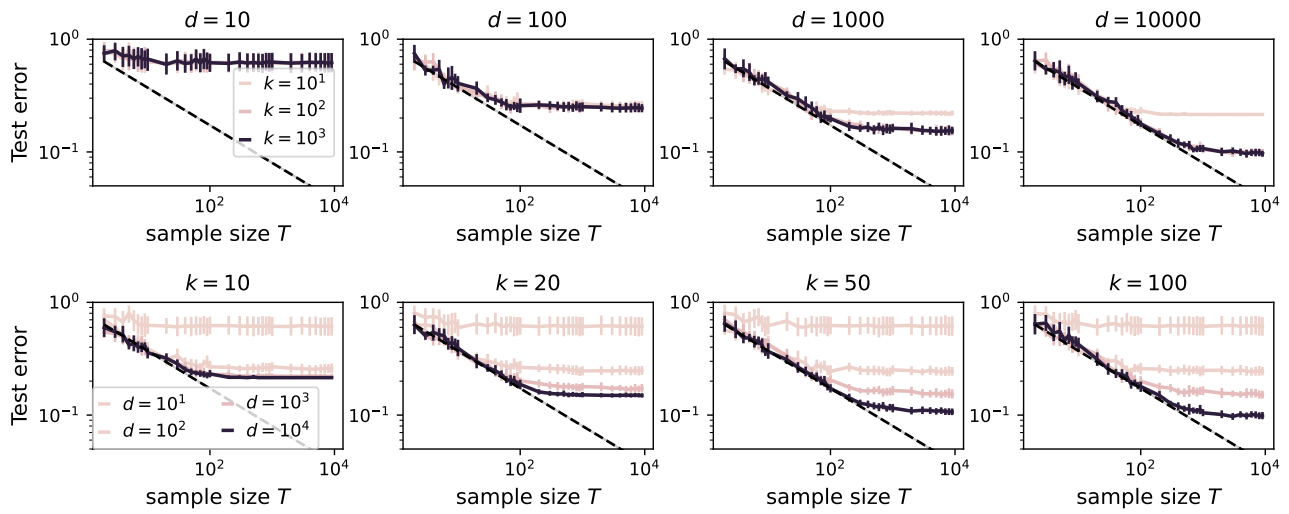

Figure 21. **Capacity-Limited Memory Models.** Empirical confirmation of the Triplet Scaling Law established in Theorem 5.1

E. Proof and Illustration of Triplet Scaling Law (Theorem 5.1)

For any i, on average it takes $1/p_i$ iid samples from p to see the context i at least once. The effect of tail-cutting at rank k is effectively to replace the sample size T by $\min(T, T_k)$, where $T_k = \max\{1/p_i \mid i \in [k]\}$. In the case where $p = Zipf(\beta)$, we have $T_k = 1/p_k \asymp k^\beta$. On other hand the model (18) proposed in (Cabannes et al., 2023) on Zipf data, the test error writes

$$E_{test} \asymp T^{-c} + d^{-c_q}, \tag{41}$$

where $c := 1 - 1/\beta \in (0, 1)$ and the exponent $c_q \in (0, \infty)$ depends on β and the algorithm q used to update the embeddings in the memory matrix M_T in (18). We deduce that tail-cutting at rank k changes the test error to

$$E_{test} \asymp \min(T, T_k)^{-c} + d^{-c_q} \asymp T^{-c} + k^{-\beta c} + d^{-c_q},$$

as claimed. □

Figure 21 confirms the Triplet Scaling Law.

F. Details and Results from the Autoregressive Bigram model with Perplexity

We showcase experiments in the autoregressive bigram model with perplexity loss. We generate sequences of length 100. Figures 22, Figure 23 and 24 aim to reproduce the "paired bigram" Figure 12 in this setting, adding a top p^{inf} mechanism and a temperature mechanism. Figure 26, Figure 26 and Figure 27 regenerates the setting of Figure 6 with the same top p^{inf} and temperature.

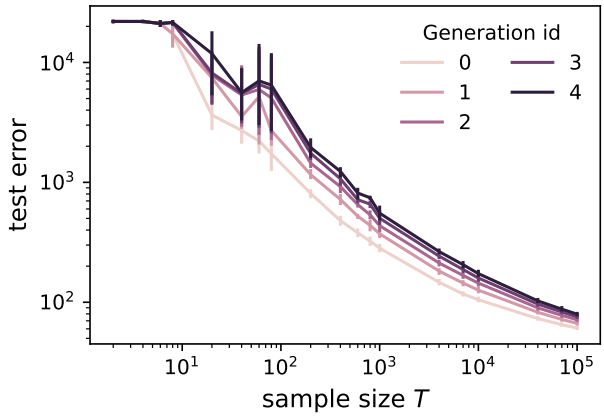

Figure 22. **Autoregressive Bigram Model with Perplexity Loss.** Empirical confirmation of Theorem 2.4 for autoregressive data with top-p^{inf} =1, Temperature τ 1. Each sequence data have length 100. Same setting as Figure 12. $\beta = 3/2$.

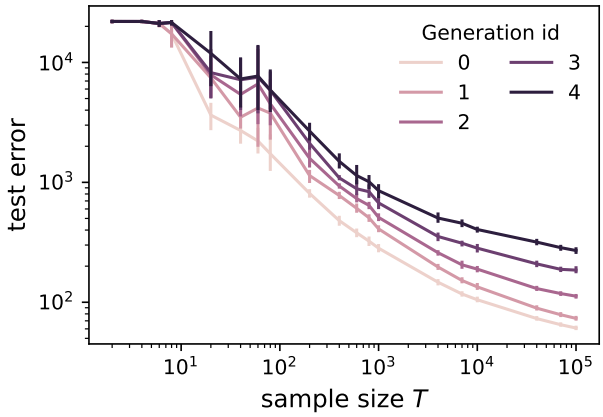

Figure 23. **Autoregressive Bigram Model with Perplexity Loss.** Empirical confirmation of Theorem 2.4 for autoregressive data with top-p^{inf} =0.9, Temperature τ 1. Each sequence data have length 100. $\beta = 3/2$.

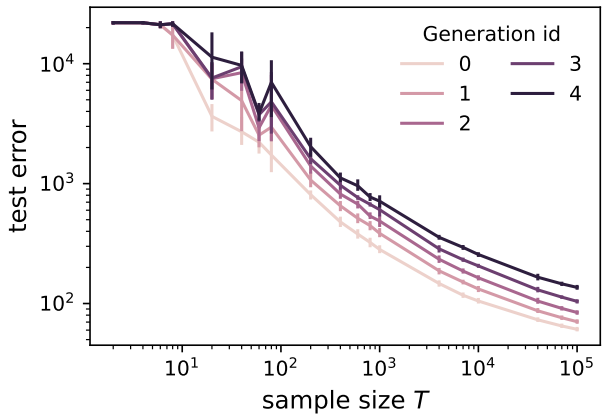

Figure 24. **Autoregressive Bigram Model with Perplexity Loss.** Empirical confirmation of Theorem 2.4 for autoregressive data with top-p^{inf} =1, Temperature τ =0.9. Each sequence data have length 100. Same setting as Figure 12. $\beta = 3/2$.

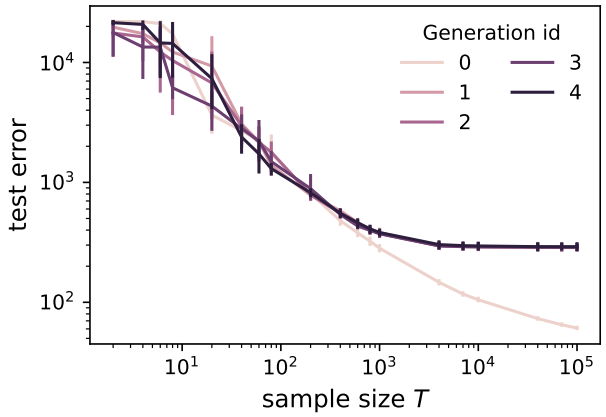

Figure 25. **Autoregressive Bigram Model with Perplexity Loss.** Each sequence data have length 100. Initial model trained on $T_0 = 10,000$ samples. It generates T samples for Gen 1. Starting from Gen 2 models are trained on data generated by the most powerful model from the previous generation. Top-p^{inf} =1, temperature τ =1, $\beta = 3/2$.

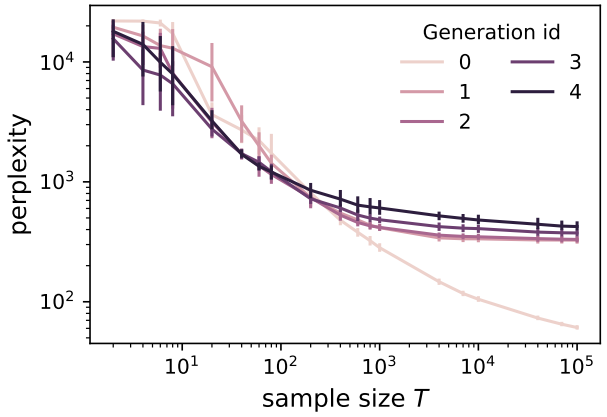

Figure 26. **Autoregressive Bigram Model with Perplexity Loss.** Each sequence data have length 100. Same setting as Figure 25. Top-p^{inf} =0.9, temperature τ =1, $\beta = 3/2$.

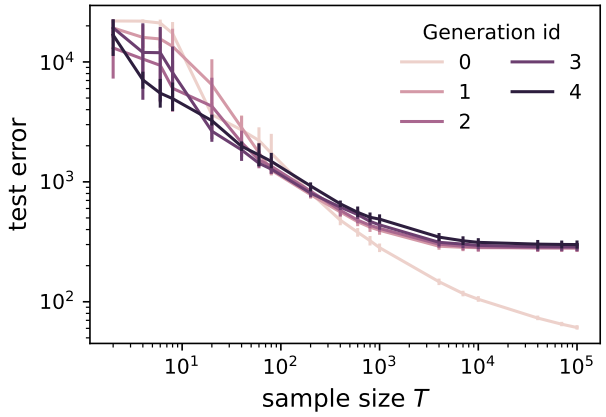

Figure 27. **Autoregressive Bigram Model with Perplexity Loss.** Each sequence data have length 100. Same setting as Figure 25. Top-p^{inf} =1, temperature τ =0.9, $\beta = 3/2$.